Potential Research Gaps

Based on the Literature Matrix

After comparing the papers in the literature matrix, I identified five possible research gaps:

Gap 1 Connecting uncertainty to the decision to answer clarify or abstain

Evidence from papers: Papers 1 to 6 study different ways of measuring or expressing uncertainty in language models, but they do not use that uncertainty to choose a conversational action. Paper 10 mainly treats abstention as refusing to answer. Papers 14, 18, and 19 study when a model should ask a clarifying question, but they do not combine this decision with calibrated uncertainty from the language model or the speech recognition system. Papers 25 and 26 study voice assistants, but not uncertainty-based clarification.

Why it matters: A voice agent needs to decide whether it has enough information to answer, whether it should ask the user a question, or whether it should avoid answering. A poor decision may lead to a confident wrong answer, too many unnecessary questions, or too many refusals.

What existing work has already tried: Previous studies have worked on confidence calibration, verbalized confidence, selective answering, abstention, clarification-question generation, and voice-agent evaluation. However, these areas are mostly studied separately.

What remains unanswered: It is still not clear how semantic uncertainty, ambiguity in the user's intention, and uncertainty from speech recognition should be combined to select the best action. It is also unclear when asking a question is more useful than answering or abstaining.

Possible research question: How can a voice agent combine semantic, intent, and speech-recognition uncertainty to decide whether to answer, ask a clarifying question, or abstain?

Gap 2 Handling speech specific ambiguity and recognition errors

Evidence from papers: The clarification studies in Papers 11 to 21 are mainly text-based. Paper 22 identifies speech recognition and confidence-based clarification as future work. Paper 24 shows that speech recognition can still produce hallucinations, repeated text, or missing words. Papers 26 and 27 evaluate spoken systems, but they do not study how the system should ask for clarification when the speech input is uncertain. Paper 28 also does not evaluate raw audio outputs directly.

Why it matters: In a spoken conversation, uncertainty may come from noise, accent, overlapping speech, or a recognition mistake. This is different from a request that is correctly recognized but does not contain enough information. The agent should not respond to both situations in the same way.

What existing work has already tried: Speech studies have improved recognition robustness and tested different acoustic conditions. Clarification studies have examined when a language model should ask a question. The reviewed papers do not fully connect these two areas in one end-to-end voice system.

What remains unanswered: It remains unclear which signals can help the agent distinguish between a misheard utterance and an ambiguous request. It is also unclear how the wording of the clarification question should change depending on the source of the uncertainty.

Possible research question: How do noise, accent, and speech-recognition errors affect a voice agent's clarification decisions, and can the agent ask a question that directly addresses the source of the uncertainty?

Gap 3 Limited evaluation with real and diverse users

Evidence from papers: Paper 13 uses an oracle language model instead of live users. Paper 17 trains and evaluates with simulated personas, and Paper 18 also depends on benchmark labels and language-model simulation. Papers 20 and 21 focus on benchmark performance. Paper 16 includes a human study, but the authors mention that the sample is small and limited to English-speaking participants from the United States.

Why it matters: A clarification question that appears useful to a simulated user may feel repetitive, confusing, or unnecessary to a real person. In voice interaction, timing, interruption, accent, privacy, and user trust may also affect the result.

What existing work has already tried: Existing studies use synthetic personas, oracle models, crowdsourced datasets, and a small number of human participants. These approaches make experiments easier to repeat, but they provide limited evidence about how the systems perform with real users in realistic situations.

What remains unanswered: We still do not know whether improvements measured with simulated users transfer to people with different accents, backgrounds, needs, and levels of technical experience. The effect on trust and perceived helpfulness also needs more study.

Possible research question: Do clarification policies trained using simulated users improve task success, efficiency, and user trust when they are tested with diverse real users in spoken conversations?

Gap 4 Clarification across longer multi turn conversation

Evidence from papers: Paper 19 focuses on one-turn or two-turn interactions and allows only one clarifying question. Paper 14 mentions multi-turn interaction as future work. Paper 27 evaluates only single-turn dialogue, while Paper 28 identifies longer conversations as an area for future research. Paper 25 supports real-time spoken interaction, but it does not study uncertainty or clarification decisions.

Why it matters: The user's answer to a clarifying question may still be incomplete, contradictory, or incorrectly recognized. Real spoken conversations can also include interruptions and corrections. A system that can ask only one question may stop too early, while a system that asks too many questions may frustrate the user.

What existing work has already tried: Previous work has studied double-turn preference training, action-based dialogue training, and real-time speech generation. These studies address useful parts of the problem, but they do not provide a complete adaptive policy for longer spoken clarification.

What remains unanswered: It is not clear how the agent should update its uncertainty after each response, decide whether another question is needed, and determine when it has enough information to answer.

Possible research question: How can a voice agent update its uncertainty across multiple clarification turns and decide when to ask again, answer, or abstain?

Gap 5 No single evaluation method for the complete interaction

Evidence from papers: Papers 1 to 7 use calibration measures such as ECE, Brier score, NLL, and AUROC. Papers 11 to 21 use measures such as answer F1, action accuracy, win rate, or task success. Papers 24 to 28 focus more on speech recognition, robustness, and voice capability. Paper 7 notes the need for broader empirical benchmarking, and Paper 28 notes that its evaluation does not operate directly on raw audio outputs.

Why it matters: A system may have well-calibrated confidence but still ask an unhelpful question. It may also resolve the correct meaning but take too many turns or respond too slowly. Looking at only one type of metric may hide problems in the overall user experience.

What existing work has already tried: The literature includes strong metrics and benchmarks for individual components. Some studies also measure user effort or the number of turns. However, the reviewed papers do not evaluate calibrated action choice, clarification quality, speech robustness, task success, latency, and user effort together.

What remains unanswered: There is no clear shared evaluation method for deciding whether an uncertainty-aware voice agent improves the complete interaction without increasing delay or asking unnecessary questions.

Possible research question: How can an end-to-end evaluation framework measure the usefulness of uncertainty-aware clarification in voice agents while considering task success, user effort, speech errors, and latency?

Initial Direction

At this stage, the first gap seems to be the most connected to the overall topic because it brings together calibration, abstention, clarification, and voice interaction.